

Customer Churn Prediction & Retention Intelligence System

Final Project Report

End-to-End Machine Learning Project for Online Fashion Store Customer Retention

Prepared For	Sarah Ahmed - Online Fashion Store
Prepared By	Muhammad Huzaifa
Project Type	Customer Churn Prediction, Risk Segmentation, Streamlit App
Primary Objective	Identify customers likely to stop purchasing and support targeted retention campaigns
Report Scope	Business understanding, EDA, modeling, tuning, deployment-ready application, and development transparency
Final Status The project has progressed from raw customer data to a complete machine learning workflow with modular Python scripts, a Streamlit dashboard, CSV validation, prediction output, and deployment readiness.	

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1. Executive Summary

This report documents the complete end-to-end development of a customer churn prediction solution for Sarah Ahmed's online fashion store. The project began with business understanding and exploratory data analysis, then progressed into feature engineering, model training, hyperparameter tuning, final model selection, customer risk segmentation, modular Python scripts, and a Streamlit dashboard for business use.

The final solution predicts whether a customer is likely to churn and assigns a business-friendly risk level: Low Risk, Medium Risk, or High Risk. The output can help the marketing team prioritize retention campaigns instead of sending generic discounts to all customers.

Executive Recommendation		
Use the model as a monthly retention intelligence tool. Upload the latest customer file, generate churn probabilities, prioritize High Risk customers, and track retention campaign outcomes over time.		
Project Component	Completion Status	Business Value
Business Understanding	Completed	Aligned the ML solution with customer retention and marketing ROI goals.
EDA and Data Cleaning	Completed	Identified duplicates, invalid dates, missing engagement values, and behavioral churn signals.
Model Training	Completed	Compared multiple classifiers and selected the best-performing baseline model.
Hyperparameter Tuning	Completed	Validated that the baseline Gradient Boosting model was already strong.
Prediction Report	Completed	Generated churn probability, risk level, and customer-level output.
Python Scripts	Completed	Converted notebook logic into reusable project modules.
Streamlit App	Completed	Created an interactive dashboard for uploading customer data and downloading predictions.
CSV Validation	Completed	Added safeguards so incorrect uploaded files show friendly validation errors.
Deployment	Ready	Recommended Streamlit Community Cloud for free public portfolio deployment.

2. Business Understanding

Sarah Ahmed operates an online fashion store and wants to reduce customer loss by identifying customers who are likely to stop purchasing. Before this project, marketing campaigns were broadly applied to many customers, which could waste discounts on customers who were not actually at risk.

2.1 Business Objectives

- Identify customers with a high probability of churn.
- Improve retention campaign targeting.
- Reduce unnecessary discount spending.
- Support monthly customer prediction reports.
- Help increase repeat purchases and customer lifetime value.

2.2 Machine Learning Problem Framing

The business problem was framed as a binary classification problem: predict whether a customer is likely to churn or remain active. Because the dataset did not include a direct historical churn label, a business rule was used to define churn based on purchase inactivity.

3. Dataset Overview

The dataset contains approximately 50,000 customer records from the online fashion store. Each row represents a customer record with demographic, purchasing, engagement, and service interaction information.

Category	Columns
Customer Identity	Customer_ID, Name
Demographics	Age, Gender, Location
Account Information	Registration_Date
Purchase Behavior	Last_Purchase_Date, Purchase_Frequency, Total_Spending, Average_Order_Value
Engagement Behavior	Website_Visits, Email_Open_Rate
Service Interaction	Customer_Support_Tickets

3.1 Data Quality Overview

The dataset was usable, but it contained several realistic data quality issues. These were documented and handled before model training.

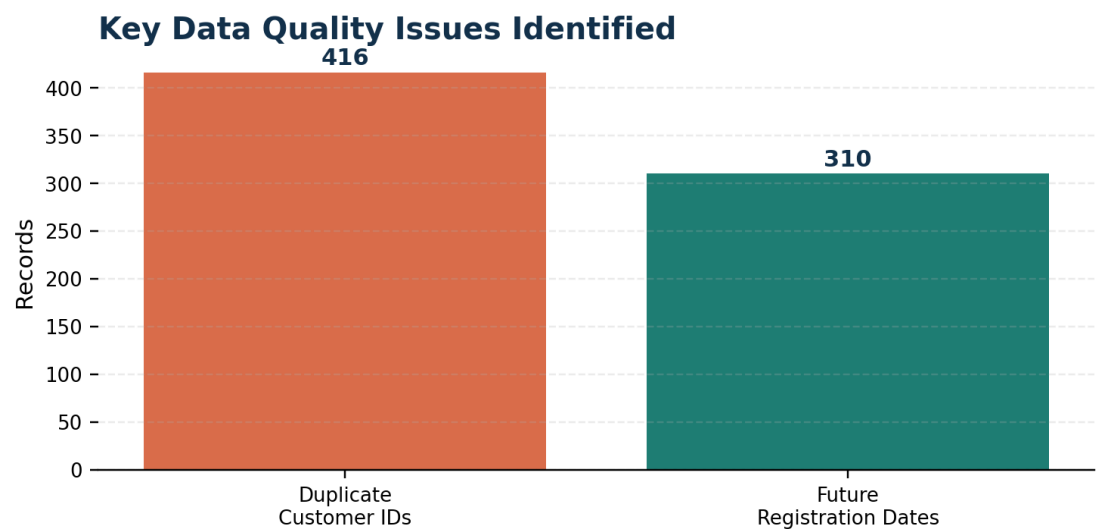


Figure 1. Key data quality issues identified during EDA.

- 416 duplicate customer records were identified through Customer_ID.
- 310 future Registration_Date values were identified and excluded from tenure calculations.
- Email_Open_Rate contained missing values and required imputation plus a missing-value indicator.
- Location values required standardization due to inconsistent naming formats.
- Customer_ID and Name were retained for reporting but excluded from model training.

4. Exploratory Data Analysis Summary

The EDA phase focused on understanding customer behavior and identifying which features showed meaningful relationships with churn risk. The strongest patterns came from customer behavior and engagement rather than demographic variables.

4.1 Demographic Features

Age, gender, and location were analyzed to understand customer distribution. Age ranged from 18 to 63, with an average age of approximately 31 years. The customer base was mostly female, with approximately 69% female and 31% male customers. However, gender-based purchase behavior and engagement were very similar.

Feature	Finding	Modeling Decision
Age	Customers mostly concentrated between 25 and 36 years.	Retained as a supporting feature.
Gender	Female and male customers showed very similar purchase and engagement behavior.	Retained as a supporting feature.
Location	Most customers were concentrated in major cities; behavior differences were minor.	Standardized and retained as a supporting feature.

4.2 Purchase Recency

Purchase recency was one of the strongest behavioral signals observed during EDA. Customers who had not purchased recently showed lower frequency, lower spending, lower website activity, lower email engagement, and higher support ticket activity.

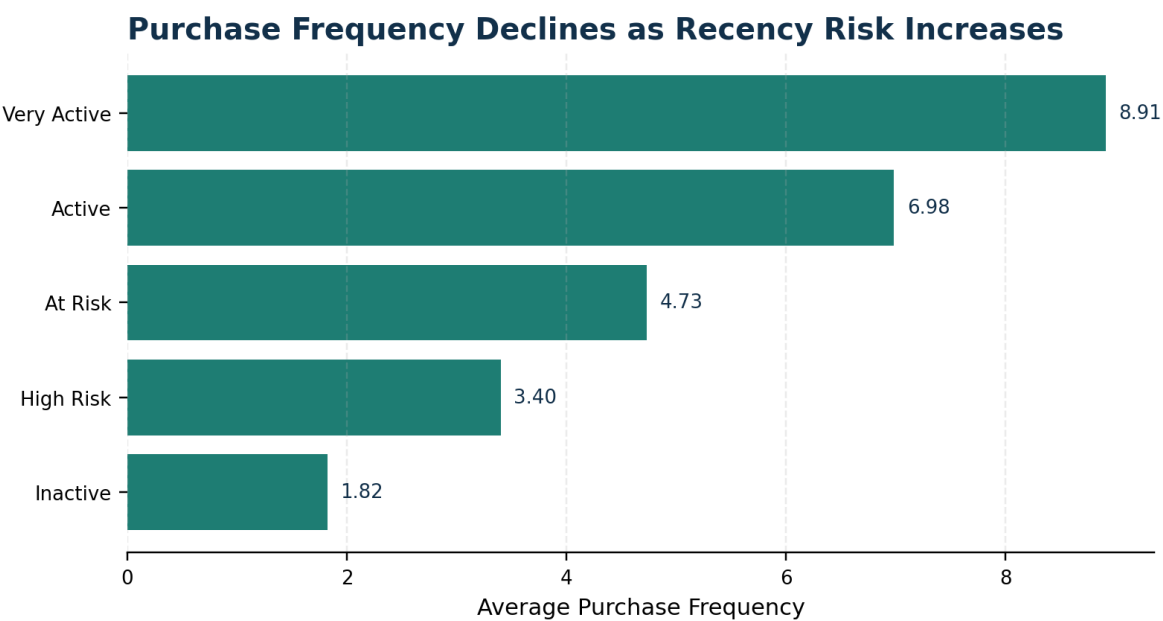


Figure 2. Purchase frequency declines as customers move from Very Active to Inactive recency groups.

Recency Group	Purchase Frequency	Total Spending	Website Visits	Email Open Rate	Support Tickets
Very Active	8.91	608.79	58.24	43.68%	1.24
Active	6.98	478.34	45.97	38.09%	1.38
At Risk	4.73	329.33	31.98	30.82%	1.78
High Risk	3.40	228.63	23.07	25.70%	2.09
Inactive	1.82	123.15	8.35	13.37%	3.13

4.3 Website Engagement

Website_Visits showed a strong relationship with purchase activity. Customers with very high website engagement had far higher purchase frequency and spending than customers with low website activity.

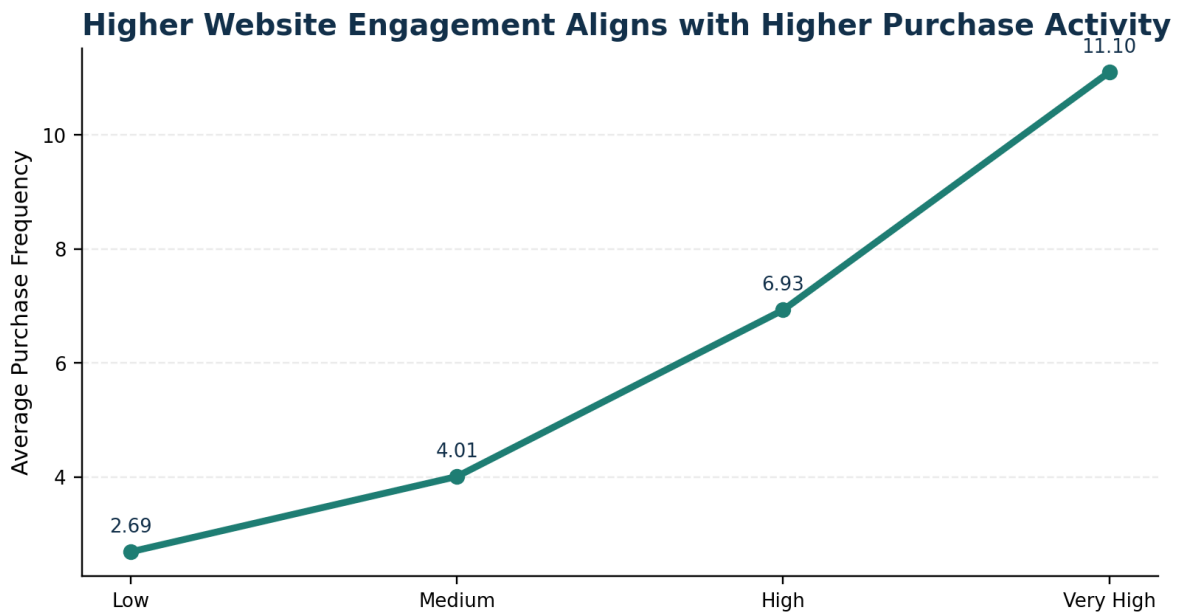


Figure 3. Higher website engagement aligns with higher purchase frequency.

Website Engagement	Purchase Frequency	Total Spending	Email Open Rate	Support Tickets	Days Since Last Purchase
Low	2.69	182.78	21.48%	2.34	126.48
Medium	4.01	275.37	28.60%	1.85	82.89
High	6.93	474.85	38.42%	1.48	48.32
Very High	11.10	756.95	49.51%	1.07	30.85

4.4 Other Behavioral Features

- **Purchase_Frequency:** Strong signal of customer loyalty and future activity. High values were kept because they may represent VIP customers.
- **Total_Spending:** Highly right-skewed, so Total_Spending_Log was created for modeling.
- **Average_Order_Value:** Moderately skewed and useful for value analysis, but not a strong standalone churn signal.
- **Email_Open_Rate:** Important engagement indicator; missing values were handled with imputation and a missing-value flag.
- **Customer_Support_Tickets:** Higher ticket activity may indicate dissatisfaction and was more common among inactive customers.

5. Data Cleaning and Feature Engineering

Data cleaning and feature engineering were performed to convert raw customer records into a reliable model-ready dataset.

Step	Action Taken	Reason
Duplicate Handling	Sorted by Last_Purchase_Date and kept the most recent record per Customer_ID.	Preserved the latest customer state rather than blindly deleting records.
Future Dates	Future Registration_Date values were treated as invalid for tenure calculation.	Prevented unrealistic negative customer tenure.
Location Standardization	Inconsistent city labels were standardized.	Improved categorical feature quality.
Customer Tenure	Created Customer_Tenure_Days from Registration_Date.	Captured how long the customer had been registered.
Purchase Recency	Created Days_Since_Last_Purchase for EDA and target creation.	Measured inactivity and recency risk.
Log Transformations	Created Total_Spending_Log and	Reduced skewness in spending-related

	Average_Order_Value_Log.	features.
Missing Indicator	Created Email_Open_Rate_Missing.	Captured whether missing engagement itself may be informative.

5.1 Final Modeling Feature Set

After EDA and leakage control, the following feature types were used for model training: demographic context, customer tenure, purchase behavior, website activity, email engagement, support activity, and transformed spending features.

```
Age
Gender
Location
Purchase_Frequency
Average_Order_Value
Website_Visits
Email_Open_Rate
Customer_Support_Tickets
Customer_Tenure_Days
Total_Spending_Log
Average_Order_Value_Log
Email_Open_Rate_Missing
```

6. Target Variable and Leakage Control

The dataset did not contain a predefined churn label. Therefore, churn was created using a business rule based on purchase inactivity.

Churn Definition

Churn = 1 if a customer has not made a purchase in the last 90 days. Churn = 0 if the customer purchased within the last 90 days.

A major modeling decision was to remove Days_Since_Last_Purchase from the training features. Although this feature was highly predictive, it was used to define the churn label. Keeping it in the model would create data leakage and produce unrealistically high results.

Column	Decision	Reason
Days_Since_Last_Purchase	Removed from training	Used to create churn target; would cause leakage.
Customer_ID	Removed from training	Identifier only, no behavioral meaning.
Name	Removed from training	Personal label, no predictive value.
EDA grouping columns	Removed from training	Created only for analysis and visualization.
Registration_Date and Last_Purchase_Date	Removed from training	Raw dates replaced by engineered features.

7. Model Training and Evaluation

Multiple classification algorithms were trained and evaluated using a train-test split with stratification. The evaluation focused on Accuracy, Precision, Recall, F1 Score, and ROC AUC.

7.1 Why Recall and F1 Score Matter

For customer churn, missing an at-risk customer can lead to lost revenue. Therefore, Recall is important because it measures how many actual churn-risk customers the model successfully identifies. F1 Score was also important because it balances Recall with Precision, preventing the business from targeting too many customers incorrectly.

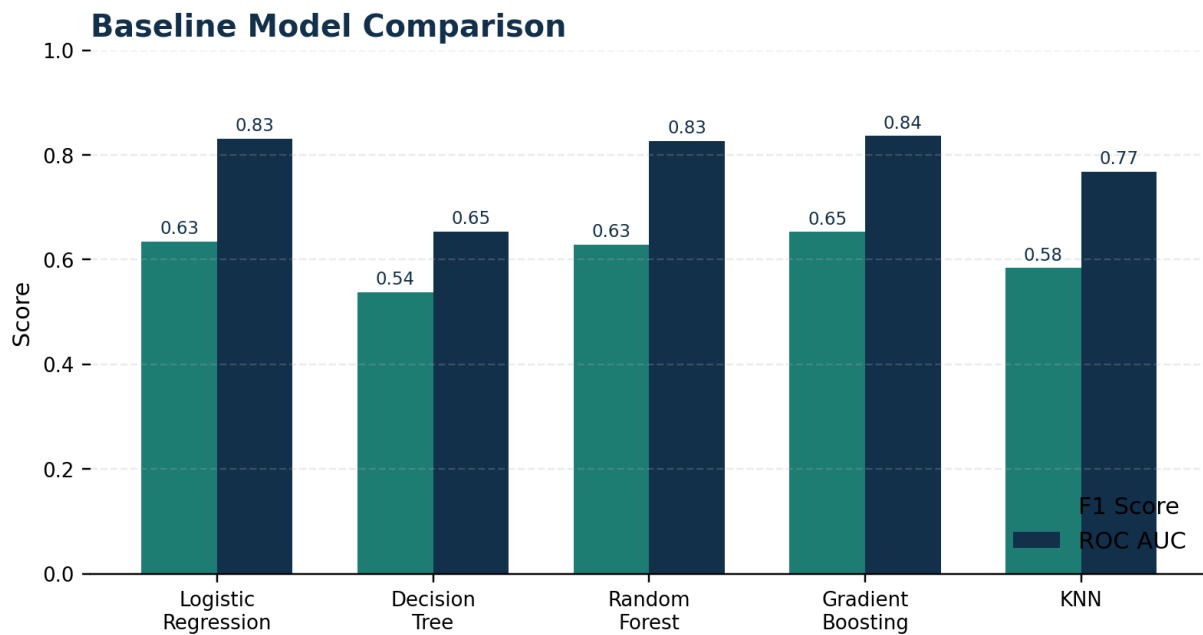


Figure 4. Baseline model comparison across F1 Score and ROC AUC.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.7676	0.6598	0.6107	0.6343	0.8308
Decision Tree	0.6909	0.5310	0.5451	0.5379	0.6539
Random Forest	0.7678	0.6655	0.5959	0.6288	0.8265
Gradient Boosting	0.7759	0.6678	0.6386	0.6529	0.8366
KNN	0.7346	0.6048	0.5649	0.5842	0.7684

Selected Baseline Model

Gradient Boosting achieved the best balance of F1 Score and ROC AUC, making it the strongest baseline model for this churn prediction problem.

7.2 Confusion Matrix Interpretation

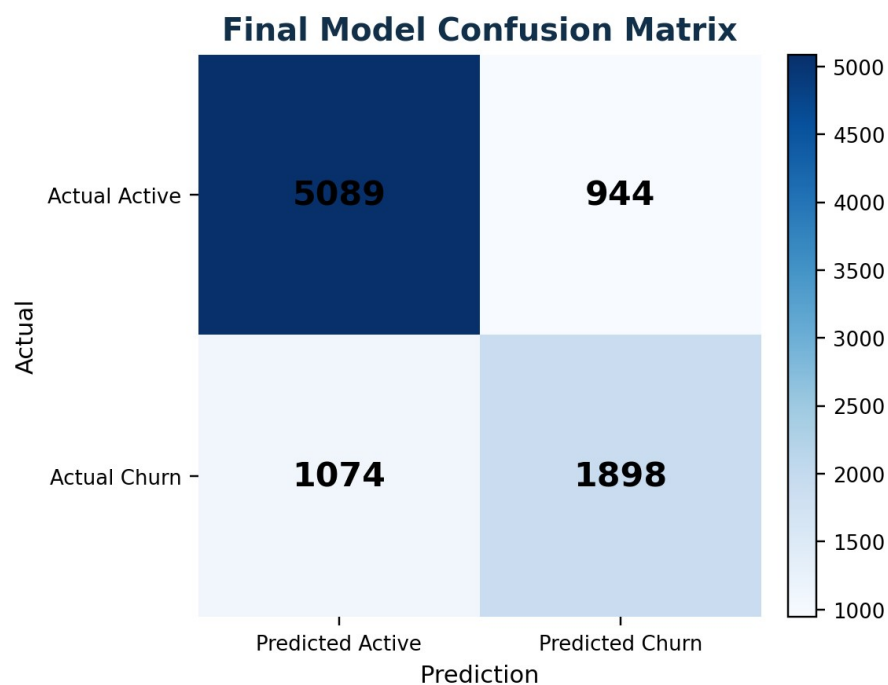


Figure 5. Confusion matrix for the final selected model.

- 5,089 active customers were correctly predicted as active.
- 1,898 churn-risk customers were correctly identified.
- 944 active customers were incorrectly flagged as churn-risk.
- 1,074 churn-risk customers were missed by the model.

8. Hyperparameter Tuning and Final Model Selection

Optuna was used to tune the Gradient Boosting model. The goal was to improve F1 Score by searching through combinations of `n_estimators`, `learning_rate`, `max_depth`, `min_samples_split`, `min_samples_leaf`, and `subsample`.

Metric	Baseline Gradient Boosting	Tuned Gradient Boosting
Accuracy	0.7759	0.7745
Precision	0.6678	0.6647
Recall	0.6386	0.6390
F1 Score	0.6529	0.6516
ROC AUC	0.8366	0.8366

Although Optuna identified a tuned parameter set, the tuned model did not meaningfully improve test performance. The baseline Gradient Boosting model was therefore selected as the final model because it performed slightly better on F1 Score and remained simpler to maintain.

Final Model Decision

Final model: GradientBoostingClassifier with the baseline configuration inside a full scikit-learn Pipeline containing preprocessing and classifier steps.

8.1 Feature Importance Insights

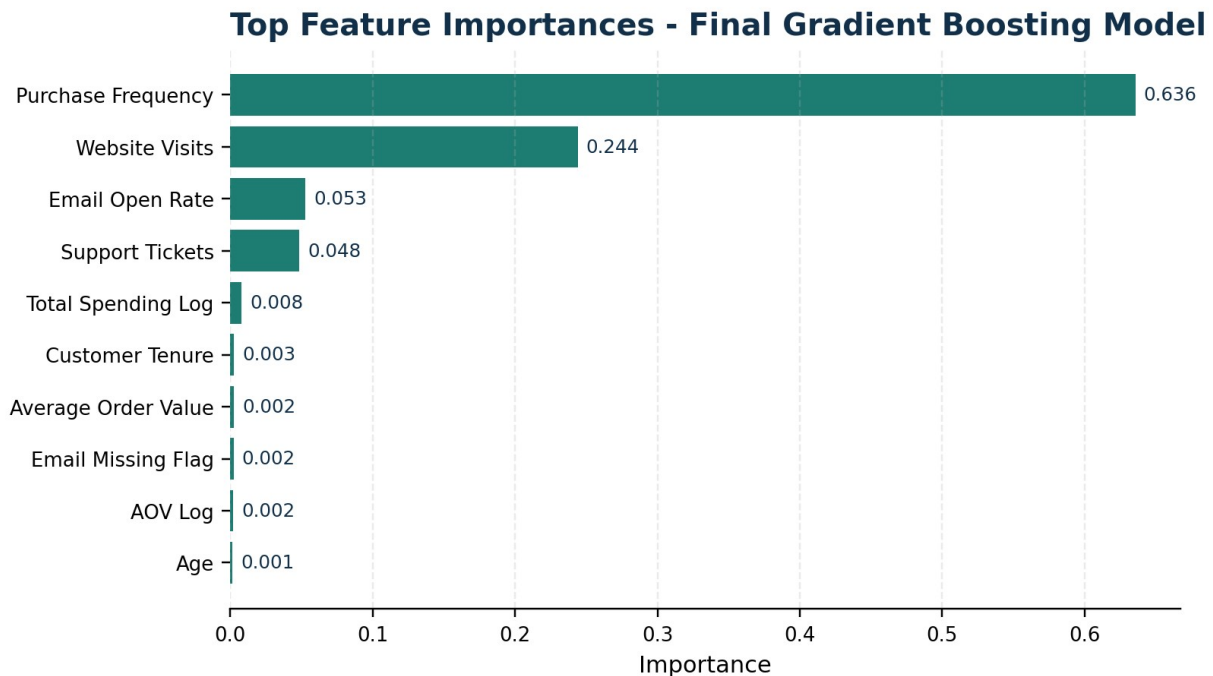


Figure 6. Top feature importances from the final Gradient Boosting model.

The model relied most heavily on `Purchase_Frequency` and `Website_Visits`. These features accounted for most of the predictive power, confirming the EDA finding that churn risk is driven mainly by behavior and engagement.

- Purchase_Frequency was the strongest predictor of churn risk.
- Website_Visits was the second strongest predictor and acted as a major engagement signal.
- Email_Open_Rate and Customer_Support_Tickets provided additional churn-risk context.
- Age and average order value had relatively low predictive influence.

9. Customer Risk Segmentation

After the final model predicts churn probability, the output is converted into simple business risk levels. This makes the prediction easier for the marketing team to use.

Risk Level	Probability Range	Suggested Business Action
Low Risk	< 30%	Continue regular engagement and standard campaigns.
Medium Risk	30% to 70%	Send promotional email or personalized reactivation offer.
High Risk	>= 70%	Prioritize immediate retention campaign and customer follow-up.

The final prediction report includes customer-level identifiers, churn prediction, churn probability, risk level, and recommended action. This output can be exported as a CSV and used by the marketing team.

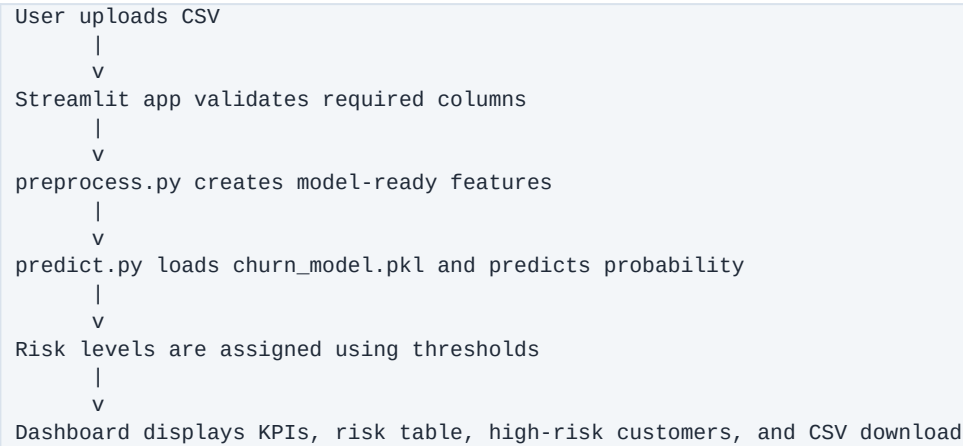
10. Technical Architecture and Codebase

The project was converted from notebooks into a modular project structure. This improves maintainability, makes the Streamlit app cleaner, and separates responsibilities across files.

<pre>Customer-Churn-Prediction/ -- data/ -- raw/ -- processed/ `-- sample/  -- notebooks/      -- 01_Business_Understanding_and_EDA.ipynb      -- 02_Model_Training_and_Evaluation.ipynb      -- 03_Hyperparameter_Tuning.ipynb     `-- 04_Customer_Prediction_and_Risk_Segmentation.ipynb -- src/ -- preprocess.py -- predict.py -- utils.py -- evaluate.py -- train.py `-- config.py  -- models/     `-- churn_model.pkl -- app/ `-- app.py  -- assets/  -- reports/  -- README.md  -- requirements.txt `-- .gitignore</pre>	
Script	Purpose
config.py	Stores file paths, app settings, random state, test size, and risk thresholds.
preprocess.py	Creates engineered features and prepares uploaded customer data for model prediction.
predict.py	Loads the saved model, predicts churn, calculates probability, and assigns risk level.
utils.py	Generates dashboard metrics, risk summaries, high-risk customer tables, and exports CSV files.

evaluate.py	Stores reusable model evaluation functions for accuracy, precision, recall, F1, ROC AUC, and confusion matrix.
train.py	Builds the preprocessing pipeline, splits data, trains the final model, and saves the model.
app.py	Runs the Streamlit dashboard for file upload, prediction, visualization, and downloads.

10.1 Data Flow



11. Streamlit Application and CSV Validation

A Streamlit dashboard was created to make the model usable for non-technical stakeholders. The app allows the user to upload a customer CSV file, run predictions, view summary metrics, inspect high-risk customers, and download the final prediction report.

11.1 Streamlit Features

- CSV file upload for customer data.
- Dataset preview before prediction.
- Customer churn prediction using the saved model pipeline.
- Churn probability generation.
- Risk level assignment: Low Risk, Medium Risk, High Risk.
- Dashboard summary cards for total customers, predicted churn, active customers, and churn rate.
- High-risk customer table for marketing prioritization.
- Downloadable prediction CSV report.

11.2 CSV Validation

CSV validation was added so the app does not fail silently or crash when the uploaded file has missing columns. The app checks whether required fields are present before running prediction.

Why CSV Validation Matters

The model only works on the same type of customer data used during training. Validation protects the app by checking that the uploaded dataset contains the required schema before prediction.

The model is reusable for the same business with new monthly data, as long as the columns and meanings remain consistent. A completely different dataset or business problem would require new preprocessing and model training.

12. Development Transparency and AI-Assisted Implementation

This section is intentionally included for transparency. During the project, Huzaifa asked for assistance with writing the modular Python scripts and Streamlit application code because the scripts contained many lines and he was still developing confidence in software engineering implementation.

The use of AI assistance was not hidden. It was used as a learning and implementation support tool while Huzaifa continued to make important project decisions, ask technical questions, review outputs, and understand the reasoning behind the workflow.

12.1 Client-Style Response to AI Assistance

Professional Assessment

Using AI to help write Python scripts or Streamlit code is not automatically disappointing. The key professional standard is whether the developer understands what the code does, can explain the data flow, can debug issues, and can justify the modeling decisions.

Throughout the project, Huzaifa asked meaningful engineering questions about duplicate handling, outlier decisions, feature leakage, model evaluation, saving the full pipeline, deployment cost, and schema validation. These questions show that the work was not simple copy-paste; it involved active decision-making and learning.

12.2 Ownership Evidence

- Questioned whether Customer_ID should be used in modeling and correctly excluded it from training.
- Investigated duplicate customer records instead of blindly dropping data.
- Identified that Days_Since_Last_Purchase created target leakage and removed it from training.
- Compared multiple model metrics instead of relying only on accuracy.
- Understood why Gradient Boosting was selected and why Optuna tuning did not improve performance.
- Asked whether the app would work on different datasets and added CSV validation accordingly.
- Recognized that scripts and Streamlit code need to be reviewed layer by layer to build full ownership.

12.3 Recommended Learning Follow-Up

The recommended next step is to review the codebase by data flow rather than line by line. Huzaifa should be able to explain: `app.py` -> `predict_customers()` -> `prepare_features()` -> `model.predict_proba()` -> risk assignment -> dashboard output.

13. Limitations and Production Considerations

- The churn label was created using a 90-day business rule because the dataset did not contain a historical churn label.
- In a real production system, churn labels should ideally be based on future behavior, such as whether a customer purchases in the next 90 days.
- The model is designed for the same type of online fashion customer data and may not generalize to unrelated datasets.
- If customer behavior changes over time, the model should be retrained and monitored for performance drift.
- The Streamlit app is suitable for portfolio and lightweight business demonstration, but enterprise deployment would require authentication, logging, monitoring, testing, and security hardening.
- Future production enhancements could include automated monthly data ingestion, campaign outcome tracking, A/B testing, and model monitoring.

14. Deployment Plan

AWS deployment was considered, but cost was a concern. For a public portfolio project, Streamlit Community Cloud is the recommended first deployment platform because it is free, simple, and provides a shareable live demo link.

Deployment Option	Best Use	Decision
Streamlit Community Cloud	Free public portfolio demo	Recommended now.
AWS EC2 / Lightsail	Cloud skills practice and production-like deployment	Optional later due to possible cost.
Render / Railway / Hugging Face Spaces	Alternative app hosting platforms	Possible alternatives if needed.

14.1 Deployment Checklist

- Push the project to a clean GitHub repository.
- Add requirements.txt with required packages only.
- Confirm models/churn_model.pkl is included or generated correctly.
- Verify app/app.py runs locally from a fresh environment.
- Deploy the GitHub repository on Streamlit Community Cloud.
- Add the live demo link to README, portfolio, LinkedIn, and resume.

15. Final Conclusion

This project successfully progressed from a business retention problem to a complete machine learning application. The final deliverable includes cleaned data, EDA insights, feature engineering, leakage-controlled model training, model evaluation, hyperparameter tuning, risk segmentation, modular Python scripts, a Streamlit dashboard, CSV validation, and deployment readiness.

The strongest churn indicators were customer behavior and engagement features, especially Purchase_Frequency and Website_Visits. The final model provides actionable churn probabilities and risk levels that can support targeted marketing decisions.

Final Assessment

The project demonstrates an end-to-end ML engineering workflow suitable for a strong portfolio project: business understanding, data analysis, modeling, software structure, app deployment readiness, and transparent AI-assisted learning.

Appendix A. Key Metrics and Tables

A.1 Final Model Performance

Metric	Final Baseline Gradient Boosting
Accuracy	0.7759
Precision	0.6678
Recall	0.6386
F1 Score	0.6529
ROC AUC	0.8366

A.2 Final Model Feature Importance

Feature	Importance
Purchase_Frequency	0.635617
Website_Visits	0.244015
Email_Open_Rate	0.052769
Customer_Support_Tickets	0.048332

Total_Spending_Log	0.007844
Customer_Tenure_Days	0.002606
Average_Order_Value	0.002345
Email_Open_Rate_Missing	0.002313
Average_Order_Value_Log	0.002073
Age	0.001444

A.3 Final Deliverables Completed

Deliverable	Status
EDA Report	Completed
Cleaned Dataset	Completed
Model Training Notebook	Completed
Hyperparameter Tuning Notebook	Completed
Prediction and Risk Segmentation Notebook	Completed
Saved Model Pipeline	Completed
Modular Python Scripts	Completed
Streamlit Dashboard	Completed
CSV Validation	Completed
Final Project Report	Completed